Yanjun Chen

yanjunc2@illinois.edu
https://chen-yanjun.com

EDUCATION

University of Illinois Urbana-Champaign, United States — Aug 2024 – Present
Ph.D. Student in Electrical and Computer Engineering
Advisor: Prof. Craig Shultz
M.S. in Electrical and Computer Engineering, August 2026

Chengdu University of Information Technology, China — Sep 2016 – Jun 2020
B.E. in Computer Science and Technology

RESEARCH EXPERIENCE

University of Illinois Urbana-Champaign, United States — Jun 2024 – Current
Research Assistant | Advisor: Prof. Craig Shultz
Research on haptic perception and actuator design for fingertip interaction.

Simon Fraser University, Canada — Oct 2023 – Apr 2024
Academic Visitor | Advisor: Prof. Xing-Dong Yang
Research on novel interactive systems using functional materials.

Institute of Software, Chinese Academy of Sciences, China — Aug 2019 – Jul 2023
Research Assistant | Advisor: Prof. Teng Han
Conducted research on haptic feedback and novel sensing for wearable and interactive systems, including compact actuators/sensors, soft interfaces, flexible circuits, and microfluidic sensing. Designed, fabricated, and evaluated haptic devices using pneumatic, electrostatic, electrotactile, and electromagnetic actuation.

PUBLICATIONS

[1] **Yanjun Chen**, Xiping Sun, Jung-Hwan Youn, and Craig Shultz, "NailBender: Exploring Nail Flexion for Haptic Input and Output," in *Proceedings of the 39th Annual ACM Symposium on User Interface Software and Technology*, 2026.

[2] **Yanjun Chen**, Xiping Sun, Jung-Hwan Youn, and Craig Shultz, "Strain2Force: Compact Nail-Based Sensor for 3DOF Contact Force Estimation," in *Proceedings of the IEEE Haptics Symposium*, 2026. *Finalist for Best Conference Technical Paper Award, Regular Paper category.*

[3] Jung-Hwan Youn, **Yanjun Chen**, Ali Israr, and Craig Shultz, "Sensitivity at the Edges of the Finger Enables Mixed Reality Force Feedback," *IEEE Transactions on Haptics*, pp. 1–11, 2026.

[4] Hechuan Zhang, Xuewei Liang, Ying Lei, **Yanjun Chen**, Zhenxuan He, Yu Zhang, Lihan Chen, Hongnan Lin, Teng Han, and Feng Tian, "Understanding the Effects of Restraining Finger Coactivation in Mid-Air Typing: from a Neuromechanical Perspective," in *Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology*, 2024.

[5] Si Chen, **Yanjun Chen**, Ji Yang, Teng Han, and Shanshan Yao, "Skin-integrated Stretchable Actuators Toward Skin-Compatible Haptic Feedback and Closed-Loop Human-Machine Interactions," *npj Flexible Electronics*, vol. 7, article 1, 2023.

[6] **Yanjun Chen**, Xuewei Liang, Si Chen, Yuwen Chen, Hongnan Lin, Hechuan Zhang, Chutian Jiang, Feng Tian, Yu Zhang, Shanshan Yao, and Teng Han, "HapTag: A Compact Actuator for Rendering Push-Button Tactility on Soft Surfaces," in *Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology*, 2022.

[7] Wei Sun, Yuwen Chen, **Yanjun Chen**, Xiaopeng Zhang, Simon Zhan, Yixin Li, Jiecheng Wu, Teng Han, Haipeng Mi, Jingxian Wang, Feng Tian, and Xing-Dong Yang, "MicroFluID: A Multi-Chip RFID Tag for Interaction Sensing Based on Microfluidic Switches," *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, vol. 6, no. 3, article 125, 2022.

[8] Chutian Jiang, **Yanjun Chen**, Mingming Fan, Liuping Wang, Luyao Shen, Nianlong Li, Wei Sun, Yu Zhang, Feng Tian, and Teng Han, "Douleur: Creating Pain Sensation with Chemical Stimulant to Enhance User Experience in Virtual Reality," *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, vol. 5, no. 2, article 89, 2021.

[9] Wei Sun, **Yanjun Chen**, Simon Zhan, Teng Han, Feng Tian, Hongan Wang, and Xing-Dong Yang, "RElectrode: A Reconfigurable Electrode for Multi-Purpose Sensing Based on Microfluidics," in *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*, 2021.

[10] Teng Han, Shubhi Bansal, Xiaochen Shi, **Yanjun Chen**, Baogang Quan, Feng Tian, Hongan Wang, and Sriram Subramanian, "HapBead: On-Skin Microfluidic Haptic Interface using Tunable Bead," in *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*, 2020.

[11] Qin Wu, Chenmei Yu, **Yanjun Chen**, Jiayu Yao, Xi Wu, Xiaolan Peng, and Teng Han, "Squeeze the Ball: Designing an Interactive Playground towards Aiding Social Activities of Children with Low-Function Autism," in *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*, 2020.

[12] Rao Xu, **Yanjun Chen**, and Qin Wu, "Face HotPot: An Interactive Projection Device for Hot Pot," in *Companion Proceedings of the 2020 Conference on Interactive Surfaces and Spaces*, 2020.

[13] **Yanjun Chen**, Chenmei Yu, Sirui Wang, and Qin Wu, "Anmoji: A Novel Body-Controlled Massage Robot," in *Companion Publication of the 2019 ACM Designing Interactive Systems Conference*, 2019.

DEMONSTRATIONS

[1] Demonstrating Differential Dual-Side Fingertip Actuation for Virtual Reality Force Feedback
Yanjun Chen, Jung-Hwan Youn, Craig Shultz
IEEE Haptics Symposium 2026, Distinguished Demonstration Award

[2] Interactive Tracking Robot Group Based on Footprint Reproduction
Yanjun Chen, Haowei Xu, Xiaoqian Xie, and Rui Deng
UIST 2019 Student Innovation Contest

[3] Anmoji: A Massage Robot Based on the Head and Shoulders Control
Yanjun Chen, Chenmei Yu, Yuwei Li, Sirui Wang, Yao Zeng, and Jingyun Yang
UIST 2018 Student Innovation Contest, Honorable Mention Award

AWARDS & SCHOLARSHIPS

Outstanding Graduate Award, Sichuan Province, China 2020
China National Scholarship, Top 0.2%, China 2017

ACADEMIC SERVICE

Reviewer: UIST 2026, CHI 2026, CHI 2025 Late-Breaking Work, UIST 2025, IMWUT 2023
Student Volunteer: IEEE Haptics Symposium 2026
Contributing Organizer: UIST 2023 Workshop on Electro-actuated Materials for Future Haptic Interfaces